Results

Results I

This section reports repository facts produced by the meta-project's live introspection pass. Test counts, module inventories, pipeline-stage counts, and figure data are injected from generated artifacts. Runtime and achieved-coverage claims are intentionally omitted when no versioned benchmark or coverage artifact backs them.

Multi-Project Pipeline Execution I

The `./run.sh` interactive orchestrator can execute the public roster through the shared DAG. The table below is a structural snapshot, not a timing benchmark: counts come from repository introspection, and the coverage column states each exemplar's declared policy floor rather than an unversioned observed percentage.

Multi-Project Pipeline Execution I

Project	Effective core stages[1]	Discovered tests	Declared project floor
`template_code_project`	8	263	90%
`template_prose_project`	8	165	90%
`template_autoresearch_project`	8	318	90%

Multi-Project Pipeline Execution I

[1]"Core-only" excludes LLM-tagged and other opt-in stages according to the YAML stage tags. A fresh run must be used to establish completion status or wall-clock performance for a particular machine and dependency set.

Infrastructure Test Suite I

Metric	Value
Test files	946+
Total tests	~12,546
Infrastructure coverage gate	$\geq$60% configured floor
Prohibited mock-framework imports	Checked by the static no-mocks gate

Infrastructure Test Suite I

Exercises Pandoc/XeLaTeX paths, steganography hashing, telemetry, YAML-driven pipeline DAG resolution, HTTPS-bound optional suites (`pytest-httpserver`), and local Ollama-gated subsets.

Infrastructure Module Inventory I

The introspection module (`template_template.core.introspection`) emits the authoritative table below—every row reflects `discover_infrastructure_modules(REPO_ROOT)`.

Infrastructure Module Inventory I

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`autoresearch`	11	✓	✓	`build_autoresearch_plan`, readiness validation CLI
`benchmark`	4	✓	✓	Template harness scoring + comparative gates
`config`	0	✓	✓	Repository defaults + hardened templates
`core`	201	✓	✓	`get_logger`, `load_config`, `TemplateError`
`docker`	0	✓	✓	Containerisation scaffolding
`doctor`	14	✓	✓	Checkout diagnose/fix/undo repairs
`documentation`	23	✓	✓	`FigureManager`, `build_api_index`, `generate_markdown_table`
`fonds`	6	✓	✓	—

Infrastructure Module Inventory I

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`formal_contracts`	5	✓	✓	—
`llm`	57	✓	✓	Ollama helpers, sanitization, review + translation pipelines
`logrotate.d`	0	✓	✓	Rotation snippets (documentation-first)
`metadata`	3	✓	✓	`normalized_repository_url`, `generate_metadata_package`
`methods`	10	✓	✓	`build_methods_orchestration_plan`, methods-stage contracts + validation
`orchestration`	14	✓	✓	`PipelineRunner`, entry point for `./run.sh`
`project`	116	✓	✓	`discover_projects`, workspace management
`prose`	9	✓	✓	Markdown readability + prose tooling

Infrastructure Module Inventory I

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`provenance`	9	✓	✓	—
`publishing`	85	✓	✓	Zenodo, executable bundle, archival targets
`reference`	16	✓	✓	BibTeX models, parsers, converters
`rendering`	121	✓	✓	PDF/HTML/slide rendering, Pandoc filters
`reporting`	412	✓	✓	Coverage parsers, dashboards, executive artefacts
`research`	3	✓	✓	—
`rules`	6	✓	✓	—
`scientific`	4	✓	✓	`check_numerical_stability`, `benchmark_function`

Infrastructure Module Inventory I

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
`search`	77	✓	✓	`infrastructure.search.literature` clients + cache
`sia`	10	✓	✓	Self-Improving-AI loop: task validation, harness, metric capture
`skills`	8	✓	✓	`discover_skills`, SKILL manifest regeneration
`steganography`	13	✓	✓	Watermark overlays + hash manifests
`tools`	6	✓	✓	—
`transmission`	6	✓	✓	—
`validation`	86	✓	✓	PDF + Markdown + integrity CLIs

Infrastructure Module Inventory I

All 31 enumerated subdirectories carry Tier-1/`README.md` and Tier-2/`AGENTS.md` coverage wherever the Documentation Duality standard applies; subsets ship Tier-3 `SKILL.md` descriptors for MCP routing (`infrastructure/skills` manifest generation).

Agentic Skill Documentation Coverage I

Layer	Role
Shared agent guidance	Root `AGENTS.md`, `.agents/` skills, roles, and workflows
Structural	`AGENTS.md` directories
Skills	Optional `SKILL.md` manifests + generated `.agents/skill_manifest.json`
PAI capsule	Repository level `PAI.md` narratives

Agentic Skill Documentation Coverage I

377+ Markdown shards under `docs/` capture operational knowledge without duplicating auto-generated inventories.

DAG Reference (Declarative Executor) I

The table summarizes the 10 default full-run stages (core plus live LLM nodes) from 17 YAML declarations; core mode selects 8. Separately tagged opt-in extensions are omitted here. Entries follow dependency order rather than numeric script filenames.

DAG Reference (Declarative Executor) I

Name	Typical script / method	Responsibility	Failure semantics
Clean Output Directories	`_run_clean_outputs`	Deletes stale `output/` trees	Blocking
Environment Setup	`scripts/pipeline/stage_00_setup.py`	Validates tooling, PYTHONPATH scaffolding	Blocking
Infrastructure Tests	`scripts/pipeline/stage_01_test.py --infra-only --infra-scope pipeline-smoke`	Focused infrastructure smoke contracts	Required tests must pass
Project Tests	`scripts/pipeline/stage_01_test.py --project-only`	Project-local tests + coverage gates	Required tests and coverage must pass
Project Analysis	`scripts/pipeline/stage_02_analysis.py`	Executes `projects/<name>/scripts/*.py`	Blocking
PDF Rendering	`scripts/pipeline/stage_03_render.py`	Pandoc → XeLaTeX manuscripts	Blocking
Output Validation	`scripts/pipeline/stage_04_validate.py`	Structural PDF/markdown probes	Blocking / warnings
LLM Manuscript Review	`scripts/pipeline/stage_06_llm_review.py --reviews-only`	Local Ollama reviews	Fresh policy-bound skip; required live acceptance remains blocking

DAG Reference (Declarative Executor) I

Name	Typical script / method	Responsibility	Failure semantics
LLM Translations	`scripts/pipeline/stage_06_llm_review.py --translations-only`	Optional translations	Fresh policy-bound skip
Copy Outputs	`scripts/pipeline/stage_05_copy.py`	Mirrors deliverables → `output/<project>/`	Blocking

DAG Reference (Declarative Executor) I

`scripts/pipeline/stage_07_executive_report.py` provides separate **multi-project executive reporting**, not an additional single-project DAG node (`execute_pipeline.py`). The multi-project executor writes its aggregate summary after project execution.

Steganographic Performance Boundary I

The steganography path exposes metadata injection, SHA-256 hashing, overlay generation, and optional QR-code insertion as separately testable operations. This revision makes no latency claim because the project does not currently generate a versioned benchmark artifact recording hardware, input PDF, warm-up policy, repetitions, and dispersion. A future performance table should be compiled from such an artifact rather than copied from an ad-hoc workstation run.

Rendered via `projects/templates/template_template` (`generate_manuscript_metrics.py` → injected tokens such as 31). Architecture figures stem from `template_template.figures.architecture_viz`—font sizes constrained by §QA.

Infrastructure · 1 / 8

autore search 11 py [ARS·]	**benchmark** 4 py [ARS·]
config 0 py [ARS·]	**core** 201 py [ARS·]

Infrastructure · 2 / 8

docker 0 py [ARS·]	**doctor** 14 py [ARS·]
documen tation 23 py [ARS·]	**fonds** 6 py [ARS·]

Infrastructure · 3 / 8

formal contracts 5 py [ARS·]	**llm** 57 py [ARS·]
logrotate.d 0 py [AR··]	**metadata** 3 py [ARS·]

Infrastructure · 4 / 8

methods 10 py [ARS·]	**orchest ration** 14 py [ARS·]
project 116 py [ARS·]	**prose** 9 py [ARS·]

Infrastructure · 5 / 8

provenance 9 py [ARS·]	**publishing** 85 py [ARS·]
reference 16 py [ARS·]	**rendering** 121 py [ARS·]

Infrastructure · 6 / 8

reporting 412 py [ARS·]	**research** 3 py [ARS·]
rules 6 py [ARS·]	**scientific** 4 py [ARS·]

Infrastructure · 7 / 8

search 77 py [ARS·]	**sia** 10 py [ARS·]
skills 8 py [ARS·]	**stegano graphy** 13 py [ARS·]

Infrastructure · 8 / 8

tools

6 py
[ARS·]

transm ission

6 py
[ARS·]

validation

86 py
[ARS·]

Projects · 1 / 7

active inference 16 chapters 84 test files	**advanced literature review** 23 chapters 8 test files
autopoiesis 8 chapters 31 test files	**autore search project** 6 chapters 23 test files

Projects · 2 / 7

autosci entists 9 chapters 14 test files	**code project** 9 chapters 16 test files
data descriptor 8 chapters 7 test files	**eda notebook** 9 chapters 12 test files

Projects · 3 / 7

experiment tree 6 chapters 5 test files	**formal** 7 chapters 34 test files
gold refinement 12 chapters 25 test files	**literature meta analysis** 21 chapters 74 test files

Projects · 4 / 7

madlib 12 chapters 8 test files	**methods paper** 9 chapters 12 test files
newspaper 6 chapters 10 test files	**pitch deck** 7 chapters 14 test files

Projects · 5 / 7

pools rules tools

8 chapters
13 test files

prose project

8 chapters
13 test files

redacted report

3 chapters
7 test files

registered report

8 chapters
5 test files

Projects · 6 / 7

search project 9 chapters 33 test files	**sia** 6 chapters 16 test files
storybook 6 chapters 7 test files	**template** 21 chapters 11 test files

Projects · 7 / 7

textbook

51 chapters
13 test files

Figure 3: Live rendering of the Two-Layer Architecture from repository introspection: the infrastructure layer (top) holds the 31 enumerated infrastructure directories, each annotated with its Python file count and a four-slot documentation badge—A AGENTS.md, R README.md, S SKILL.md, P PAI.md, with · marking an absent file—so a fully documented module reads [ARSP]. A YAML DAG arrow connects it to the project

Self-Referential Analysis I

layer (bottom) of public exemplars labelled with chapter and test-file counts. The takeaway: documentation-duality coverage is near-uniform across the infrastructure, and every box was placed from the same live data the prose cites.

Stages 1–5 of 17

1	Clean Output Directories	core clean
2	Environment Setup	core
3	Infrastructure Tests	core tests
4	Project Tests	core tests
5	Project Analysis	core

Stages 6–10 of 17

6	Connector Search	science live
7	Provenance Record	provenance
8	PDF Rendering	core
9	Output Validation	core
10	LLM Manuscript Review	llm live

Stages 11–15 of 17

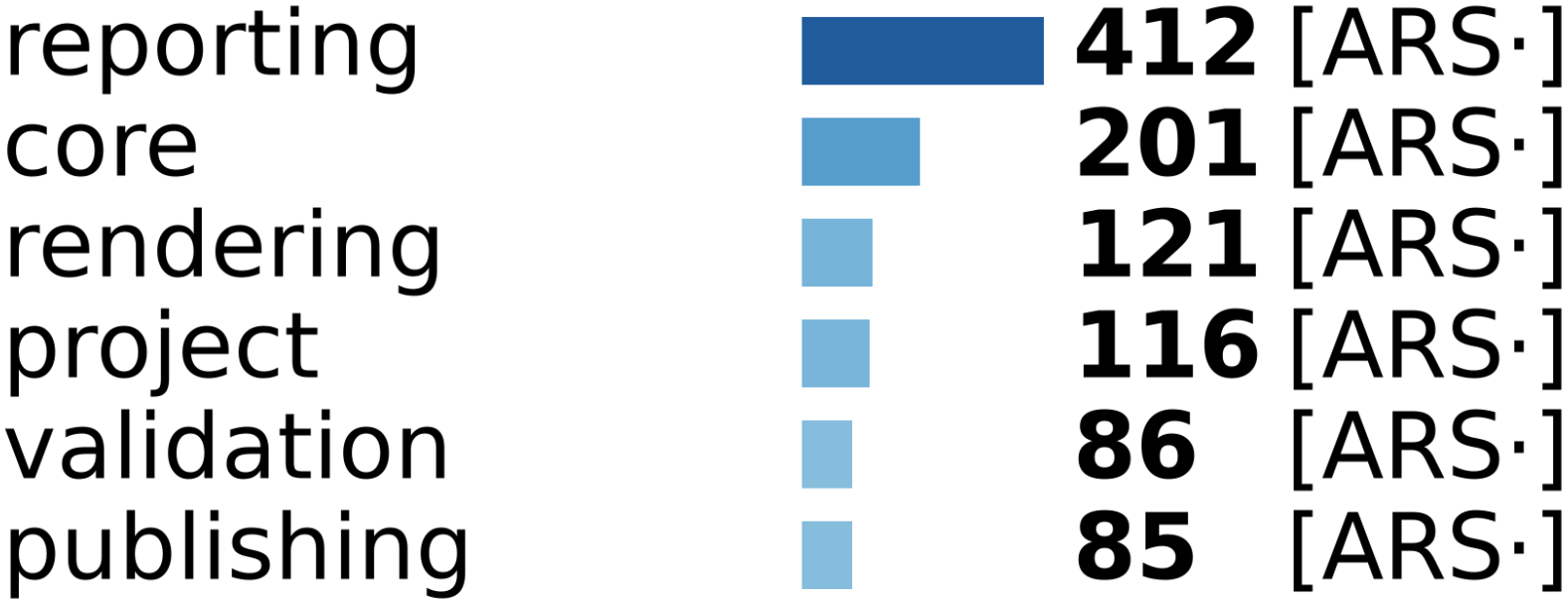
11	LLM Translations	llm live
12	Copy Outputs	core
13	Ebook Generation	core ebook
14	docxplus Export	core docxplus
15	Metadata Package	core metadata

Stages 16–17 of 17

16	Executable Bundle	bundle
17	Archival Publication	archival live

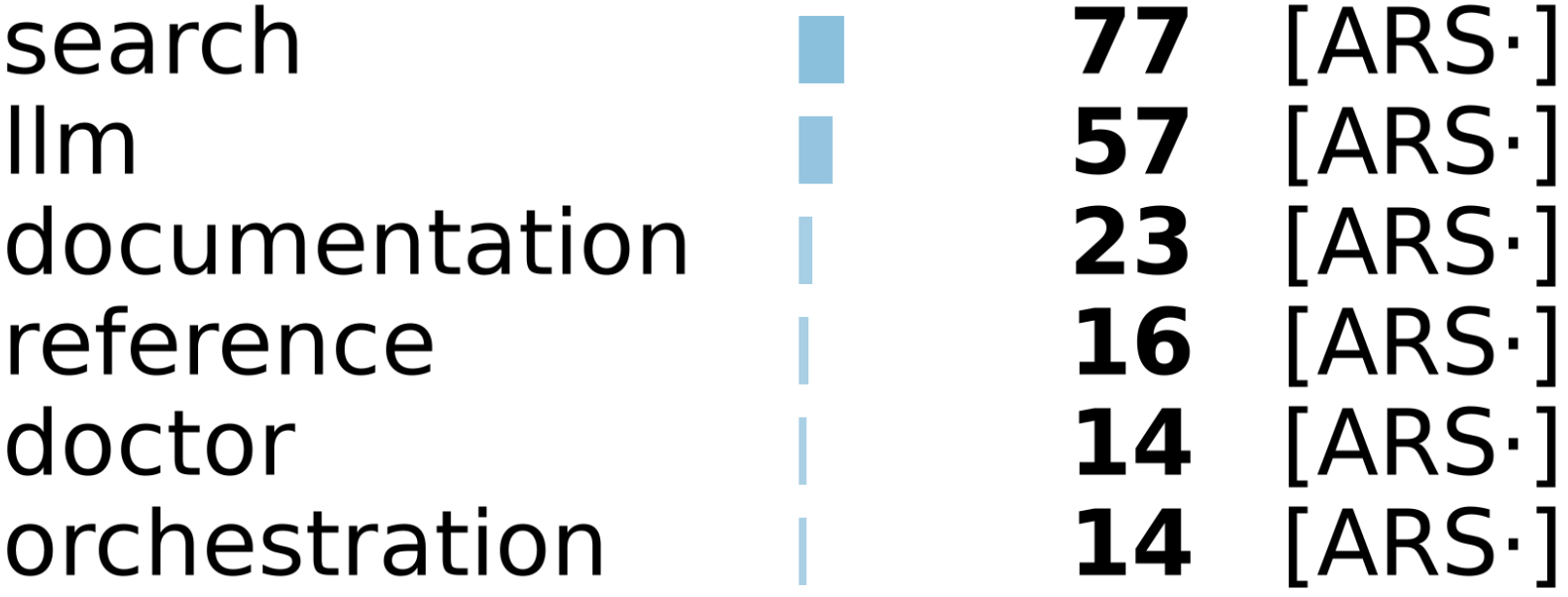
Figure 4: Declared-stage reference with all 17 YAML stages in reading order, showing complete names and tags (core, LLM, ebook, docxplus, metadata, bundle, archival). Panel order documents declarations; it does not imply a linear dependency chain.

Modules 1–6 of 31



Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Modules 7–12 of 31



Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Modules 13–18 of 31

steganography	**13**	[ARS·]
autoresearch	**11**	[ARS·]
methods	**10**	[ARS·]
sia	**10**	[ARS·]
prose	**9**	[ARS·]
provenance	**9**	[ARS·]

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Modules 19–24 of 31

skills	**8**	[ARS·]
fonds	**6**	[ARS·]
rules	**6**	[ARS·]
tools	**6**	[ARS·]
transmission	**6**	[ARS·]
formal_contracts	**5**	[ARS·]

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Modules 25–30 of 31

benchmark	**4**	[ARS·]
scientific	**4**	[ARS·]
metadata	**3**	[ARS·]
research	**3**	[ARS·]
config	**0**	[ARS·]
docker	**0**	[ARS·]

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Modules 31–31 of 31

logrotate.d **0** [AR··]

Python files · A AGENTS
R README · S SKILL · P PAI
· = absent

Figure 5: Horizontal file-count histogram of every infrastructure subdirectory, sorted largest-first.

Self-Referential Analysis I

The long tail of small, single-purpose packages beside a handful of larger ones (`core`, `validation`, `publishing`) is the visual signature of the Unix-philosophy modularity the architecture section argues for—capability concentrated where it compounds, not spread evenly by fiat.

Comparative Feature Analysis I

The comparative feature matrix below summarizes the Appendix F capability matrix.

Tool key · A–J

A template/

B Snakemake 9.x

C Nextflow 25.x

D CWL 1.2

E Quarto 1.x

F Jupyter Book 2.x

G R Markdown

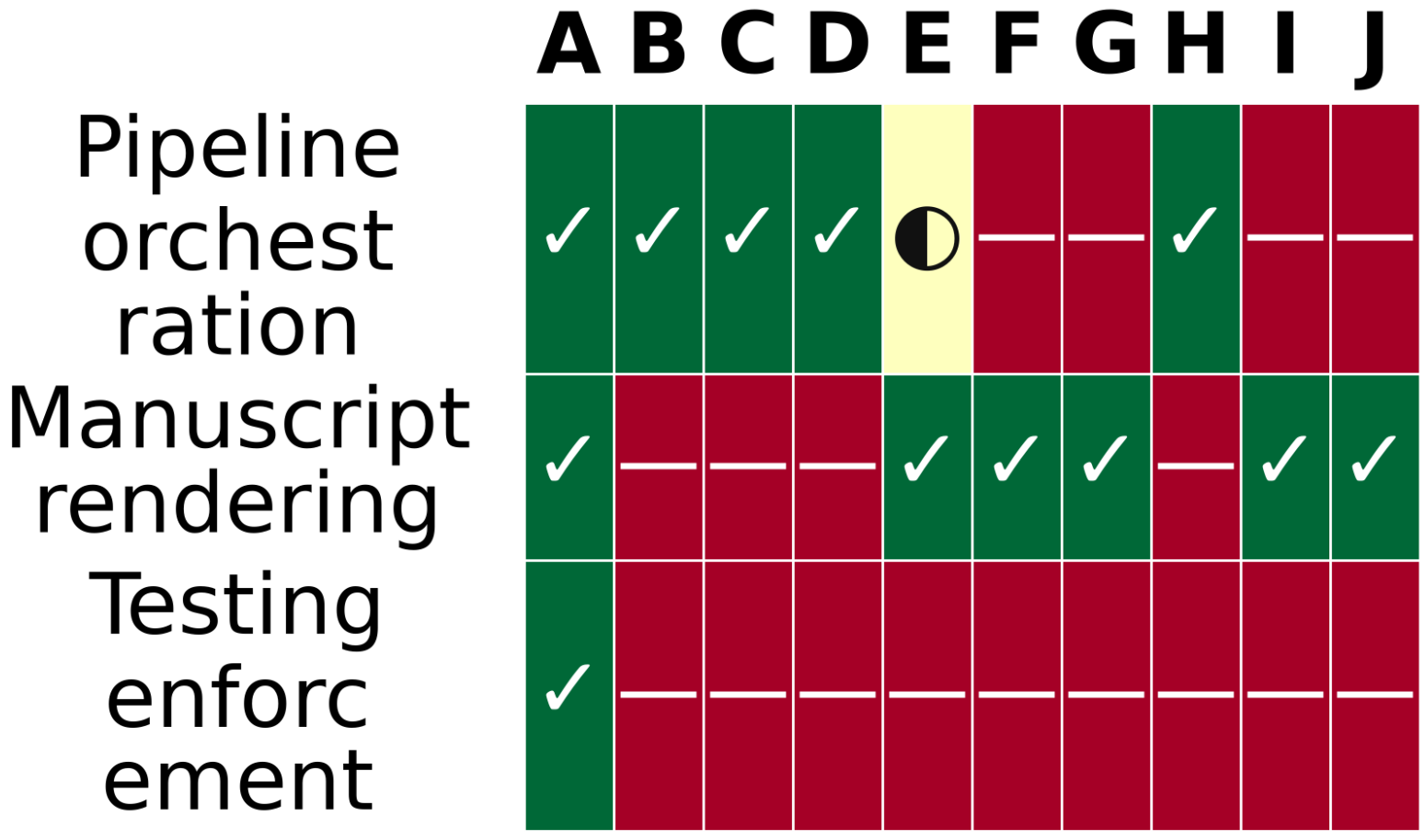
H DVC 3.x

I Overleaf (2025)

J OpenAI Prism

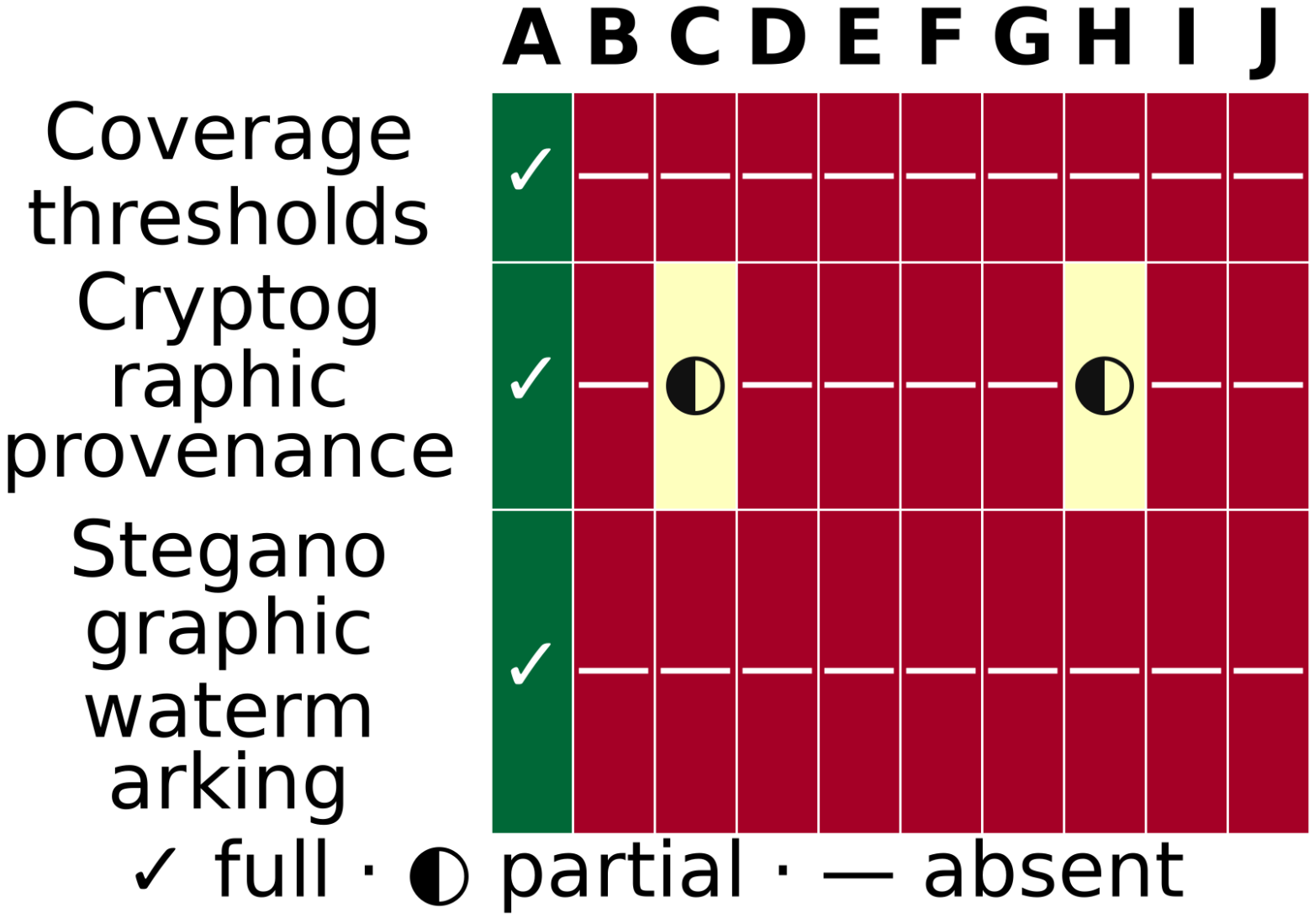
✓ full · ◐ partial · — absent

Capabilities 1–3 · tools A–J

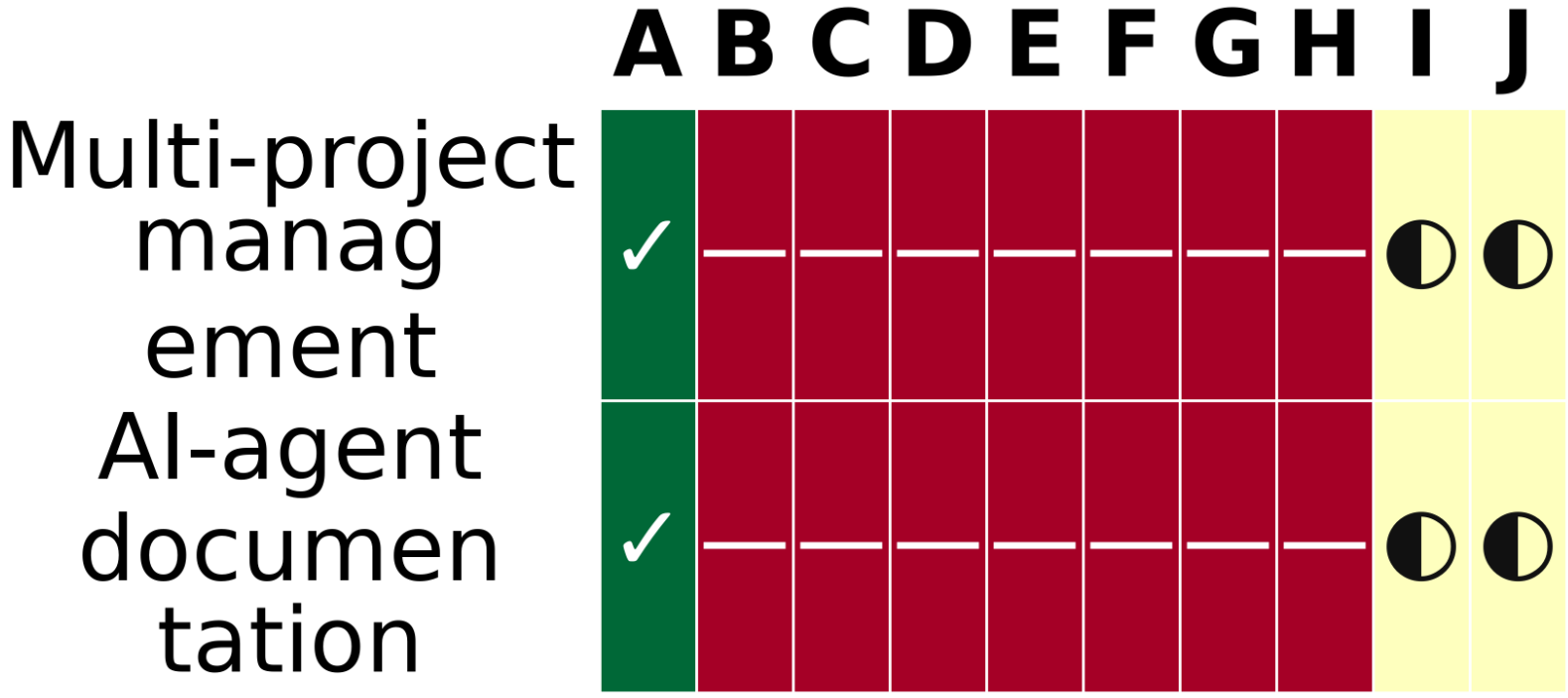


✓ full · ◐ partial · — absent

Capabilities 4–6 · tools A–J

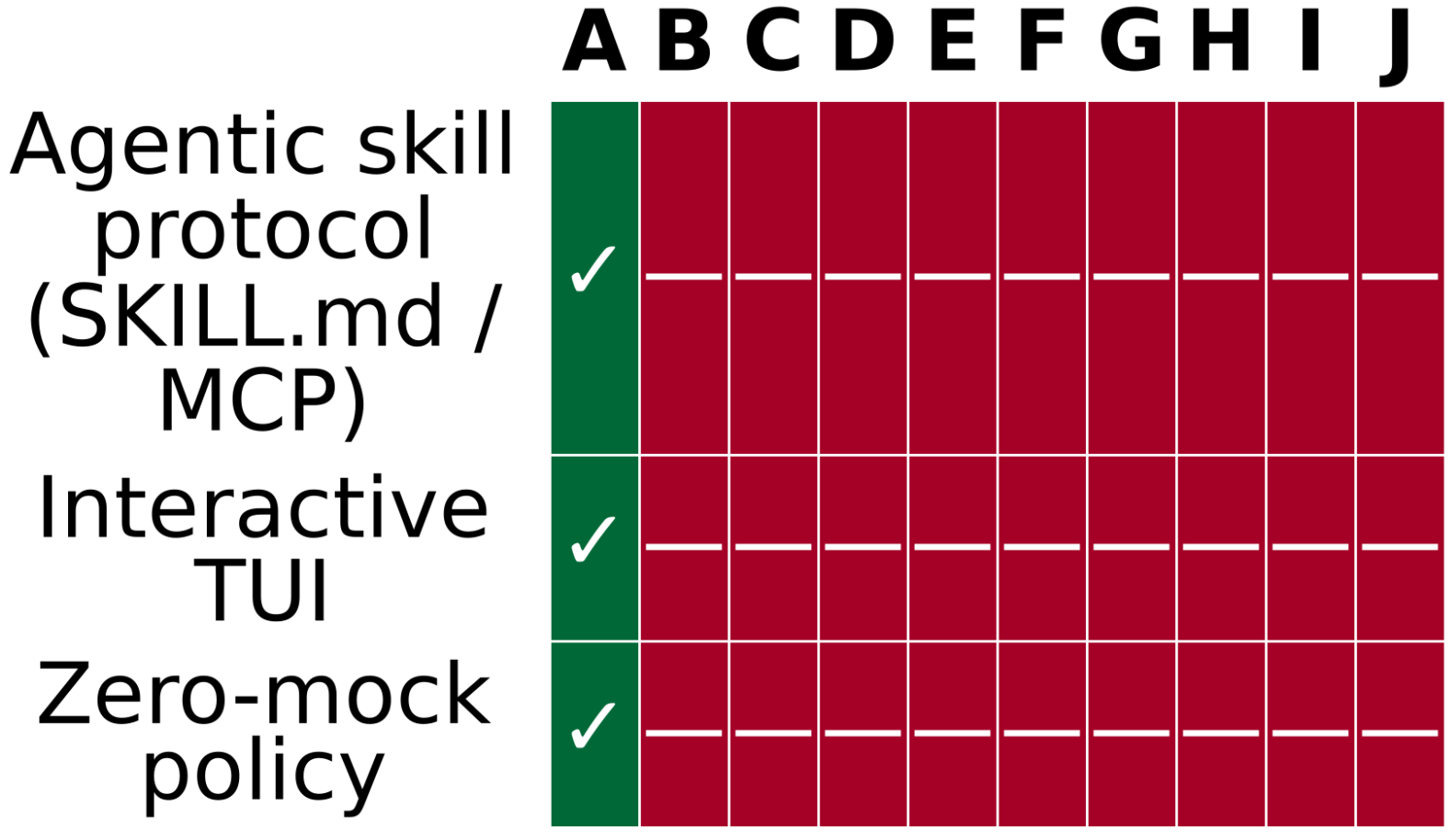


Capabilities 7–8 · tools A–J

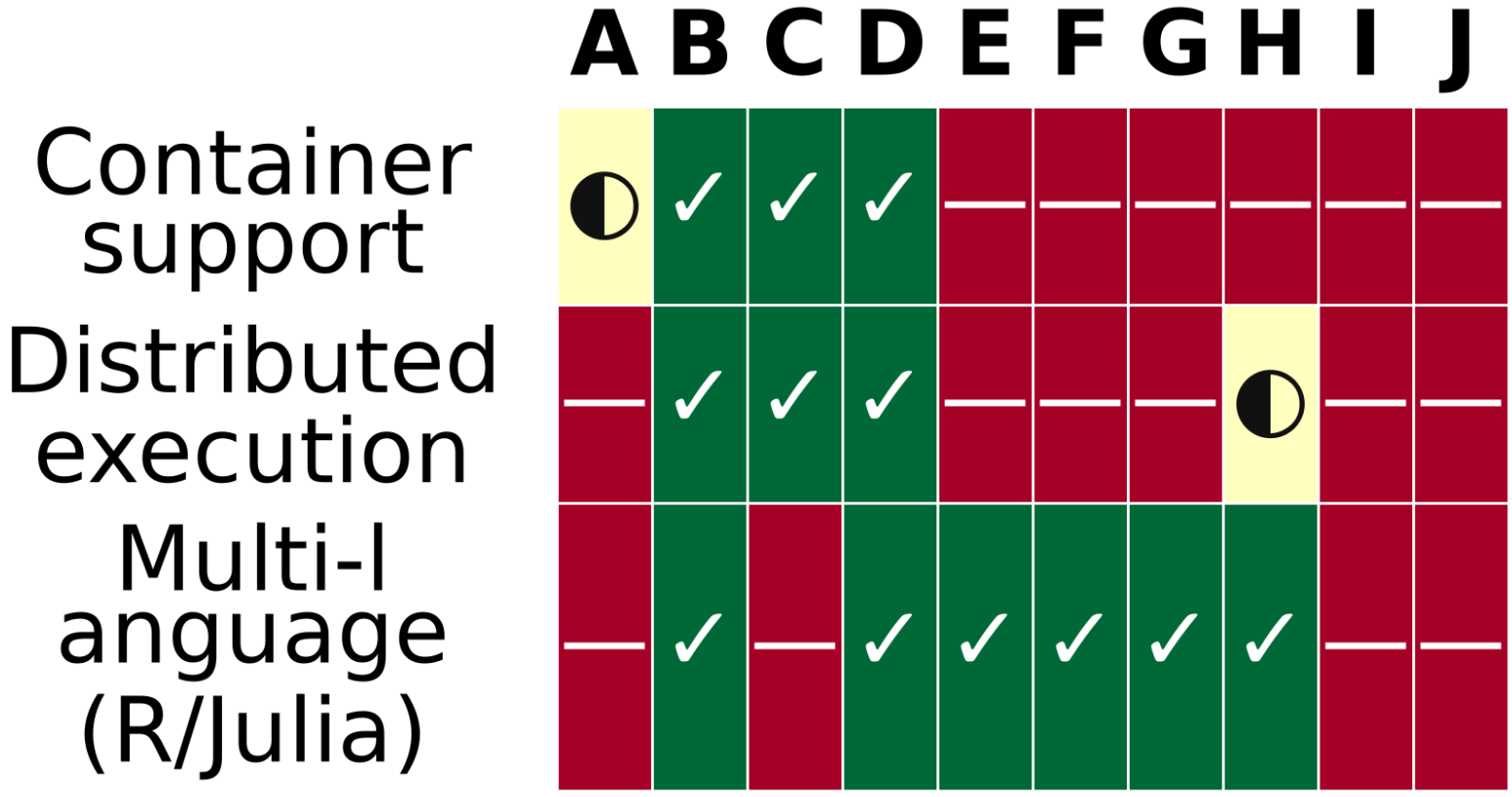


✓ full · ◐ partial · — absent

Capabilities 9–11 · tools A–J



Capabilities 12–14 · tools A–J



✓ full · ◐ partial · — absent

Figure 6: 14×10 heatmap annotated in appendix text—green ✓ full native capability, yellow ~ partial / plugin-hosted, red — unavailable. Rows group *Core Pipeline*, *Quality & Security*, then *Ecosystem*.

Comparative Feature Analysis I

[1] Nextflow lineage records exist at workflow scope, not per rendered PDF citation graph.

[2] DVC: content-addressed artifacts without native prose rendering.

[3] DVC: remote object stores (S3, GCS, Azure) without turnkey CI manuscript gates.

Test Quality Metrics I

- **Zero mocks:** repository policy bans `unittest.mock` / patching frameworks in tests.
- **Filesystem + subprocess realism:** ephemeral directories + actual CLI binaries.
- **HTTP realism:** infra suites favour `pytest-httpserver`; literature tests hit recorded fixtures.
- `template_code_project` focuses on numerical reproducibility assertions.
- `template_autoresearch_project` exercises the AutoResearch readiness planner (`infrastructure/autoresearch/`). `template_search_project` exercises literature-search workflows.